

Ankit Yadav

Khalifa University, Abu Dhabi, UAE
100062650@ku.ac.ae | yankit0511@gmail.com

Education

Doctor of Philosophy (PhD) in Astroparticle Physics

Khalifa University

- **Thesis:** *Investigation of the origin of high-energy cosmic rays using data from cosmic-ray, gamma-ray, and neutrino experiments*

Aug 2022–Present

Abu Dhabi, UAE

GPA: 3.91/4.00

BS–MS Dual Degree in Physics (Minor in Astronomy)

Indian Institute of Science Education and Research (IISER) Mohali

- **Thesis:** *Bayesian modeling of dark energy: constraining cosmological parameters with observational data*

Aug 2017–May 2022

Mohali, India

CPI: 8.4/10

Research Interests

- Astroparticle physics: origin, acceleration, and propagation of high-energy cosmic rays
- Multi-messenger astrophysics integrating cosmic rays, gamma rays, and neutrinos
- Cosmology and the large-scale universe: dark energy, structure formation, and early-universe physics
- Numerical and computational modeling of high-energy astrophysical processes
- Data-driven and machine learning approaches in astrophysical research

Research Outputs

Manuscripts in Preparation

- [1] **Yadav, A.**, Thoudam, S. *Constraints on the contribution of local sources to the observed cosmic-ray electron spectrum. In preparation.*
- [2] **Yadav, A.**, Thoudam, S., Eichmann, B., Rachen, J. P. . *Weak-shock reacceleration as a unified framework for Galactic cosmic-ray spectra. In preparation.*

Conference Proceedings

- **Yadav, A.**, Thoudam, S., Eichmann, B., Rachen, J. P. (2025). *Bayesian modeling of Galactic cosmic-ray propagation*. PoS(ICRC2025) 154. [doi:10.22323/1.501.0154](https://doi.org/10.22323/1.501.0154).

Research Experience

Diffuse Galactic Gamma-Ray Emission

Supervisor: Dr. Satyendra Thoudam

Sep 2025–Present

Khalifa University,

- Modeled diffuse Galactic γ -ray emission using cosmic-ray distributions from weak-shock reacceleration and diffusion frameworks

Propagation of Galactic Cosmic-Ray Electrons (GeV–TeV)

Supervisor: Dr. Satyendra Thoudam

Apr 2025– Sep 2025

Khalifa University,

- Investigated the $\sim$ TeV spectral break in Galactic cosmic-ray electrons; constrained maximum nearby SNR contributions and refined inverse-Compton loss modeling using updated ISRF data.
- Modeling phase completed; manuscript in preparation.

Weak-Shock Reacceleration of Galactic Cosmic Rays

Supervisor: Dr. Satyendra Thoudam

Aug 2024–Jul 2025

Khalifa University,

- Established a Bayesian inference framework for Galactic cosmic-ray propagation incorporating weak-shock reacceleration.

- Achieved unified fits to primary and secondary spectra, reproducing the observed spectral hardening and B/C ratio using AMS, CALET, CREAM, and Voyager data.
- Developed and benchmarked a hybrid Cpp–Python solver with MPI-based nested parallelism, optimized for large-scale MCMC parameter estimation on HPC clusters using SLURM.

Modeling Acceleration of Cosmic Rays

Apr 2023–Aug 2024

Supervisor: Dr. Satyendra Thoudam

Khalifa University,

- Developed a computational framework to solve complex time-dependent transport equations for diffusive shock acceleration (DSA) at supernova remnant shocks.

Model-Dependent Cosmological Parameter Fitting (Master’s Thesis)

May 2021–Apr 2022

Supervisor: Dr. Harvinder Kaur Jassal

IISER Mohali, India

- Performed Bayesian constraints on FRW dark-energy parameterizations using combined Pantheon SN Ia, Hubble, and BAO datasets.

Perihelion Precession of Mercury and Bending of Light

May 2020–Aug 2020

Supervisor: Dr. Harvinder Kaur Jassal

IISER Mohali, India

- Investigated general relativistic corrections to planetary orbits; simulated Mercury’s precession and light deflection in MATLAB.

Astrometry and Photometry of NGC 663

May 2019–Dec 2019

Supervisor: Prof. Kulinder Pal Singh

IISER Mohali, India

- Performed astrometric and photometric analysis using IRAF and SAO DS9; corrected coordinates with *AstroSat*/UVIT data.

Conferences

TeV Particle Astrophysics Conference (TeVPA 2025)

Nov 3–7, 2025

Valencia, Spain

- Poster presented: ‘Bayesian analysis of Galactic cosmic-ray transport with weak-shock reacceleration’.

International Cosmic Ray Conference (ICRC 2025)

Jul 15–24, 2025

CICG, Geneva, Switzerland

- Poster presented: ‘Bayesian modeling of cosmic-ray transport’.

Graduate Students Research Conference (GSRC 2025)

Apr 12, 2025

American University of Sharjah, UAE

- Oral presentation: ‘Bayesian modeling of Galactic cosmic-ray propagation’.

CRPropa Conference on Astroparticle Propagation 2025

Jan 6–9, 2025

Khalifa University, Abu Dhabi, UAE

- Talk: ‘Bayesian modeling of Galactic cosmic-ray propagation’.

9th UAE Graduate Students Research Conference (GSRC 2024)

Apr 20, 2024

UAE University, Al Ain, UAE

- Poster presented: ‘Numerical solution to time-dependent transport equations for cosmic rays originating from supernova remnants’.

Workshops, Schools & Training

LOFAR/SKA/IFT Workshop 2025

Jun 2–5, 2025

KIT Campus North, Germany (online)

- Focused discussions on recent progress in LOFAR and SKA, followed by sessions on IFT-based reconstruction approaches.

School on Experimental Astroparticle Physics (SEAP 2025)

Mar 9–13, 2025

TIFR Ooty, India

- Hands-on training in scintillator assembly, calibration, and shower reconstruction; lectures on cosmic rays, neutrinos, and multi-messenger astrophysics.

11th International Workshop on Very High Energy Particle Astronomy (VHEPA 2024)

Oct 17–19, 2024

Indian Institute of Technology Kanpur, India (online)

- Participated in sessions on multi-messenger astrophysics.

Workshop on Data Analysis in Cosmology and Astroparticle Physics

Aug–Sep 2021

TIH–ISI Kolkata, India

Academic Presentations & Seminars (selected)

Solving the time-dependent transport equation for cosmic rays

May 2023

Khalifa University, Abu Dhabi, UAE

Dispersion measures of fast radio bursts

Dec 2022

Khalifa University, Abu Dhabi, UAE

Gravitational waves: history, detection, and prospects

Dec 2022

Khalifa University, Abu Dhabi, UAE

Model-dependent cosmological parameter fitting

Apr & Dec 2022

Indian Institute of Science Education and Research (IISER) Mohali, India

Large-scale structures and galaxy clustering

Apr 2022

Indian Institute of Science Education and Research (IISER) Mohali, India

Skills

Computational: Numerical modeling, Bayesian inference, data analysis, and parallel computation

Programming Languages: Python, C++, Fortran, C, MATLAB

Software & Frameworks: PyMC, emcee, ROOT, Mathematica, Pybind11

High-Performance Computing: Linux-based HPC environments, SLURM scheduling, MPI, and OpenMP parallelization

Teaching & Outreach

Secretary — Outreach, Recruitment, and Communications Committee

Aug 2025–Present

Department of Physics, Khalifa University, Abu Dhabi, UAE

- Coordinating departmental outreach and recruitment initiatives, including public engagement, student events, and communications activities.

Volunteer — KU–CERN International Masterclass

Apr 10, 2025

Khalifa University, Abu Dhabi, UAE

- Guided high-school students through $Z^0 \rightarrow \ell^+ \ell^-$ reconstruction using the HYPATIA event display.
- Introduced collider-physics concepts and basic statistical techniques used in particle discovery.

Member, Organizing Committee — CRPropa Conference on Astroparticle Propagation

Jan 6–9, 2025

Department of Physics, Khalifa University, Abu Dhabi, UAE

- Assisted in organizing and coordinating conference logistics, including participant communication, scheduling, and on-site arrangements.

Primary Instructor — Introduction to Astronomy Lab

Fall 2023, Fall 2024, & Spring 2025

Khalifa University, Abu Dhabi, UAE

Member, Organizing Committee — *Astronomy Day*
Khalifa University, Abu Dhabi, UAE

Spring 2023, Fall 2024, & Spring 2025

- Organized public astronomy events as part of the departmental committee, including telescope demonstrations, planetarium sessions, and hands-on workshops for students and visitors.

Awards & Achievements

Doctoral Research / Teaching Scholarship (DRTS)
Khalifa University

Fall 2022–Present

3rd Place — AI-Driven Drug Discovery Hackathon in Abu Dhabi
(organized by Insilico Medicine)
IRENA Building, Masdar City, UAE

Nov 16, 2024

Merit-Cum-Means Fellowship for MS
Ministry of Minority Affairs, Government of India

2020–2022

GATE Qualified in Physics
Graduate Aptitude Test in Engineering (India)

Feb 2022